Problem

Problem · Drawing without lifting the pencil. An *Eulerian trail* is a walk that uses every edge of the graph exactly once.

1. Assume that a graph has an Eulerian trail whose ends are distinct. Prove that its two ends have odd degree and that all the other vertices have even degree. What can be said if the trail is closed?
2. We take the converse for granted (Euler's theorem): a connected graph all of whose vertices have even degree, except zero or two, has an Eulerian trail; if there are two vertices of odd degree, it starts at one and ends at the other.

   Consider the "house": a square $abdc$ (a and b at the bottom, c and d at the top) with its two diagonals, topped by a roof c-e-d. List the edges and the degrees.
3. Can the house be drawn without lifting the pencil or going over a line twice? If so, where must one start? Give such a trail.
4. Can it be done returning to the starting point?

Beyond the syllabus · Euler's theorem. The converse stated in question 2 (Euler's theorem) is not part of the syllabus: it is admitted and is not assessed.